

Fraud Detection via Graph Theory: A Comparative Study of Classical, Spectral, and Neural Methods

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Abstract—The application of graph-theoretic methods to financial fraud detection was examined and benchmarked against tabular anomaly detection baselines. Two real-world datasets were employed: a European credit card transaction dataset comprising 284,807 transactions, of which 0.172% were fraudulent, and a Medicare Part D provider prescribing dataset comprising 7,701 providers, of which 1.78% were labeled as fraudulent. Graphs were constructed from tabular records by using a k -nearest-neighbor cosine similarity graph for credit card transactions and a provider–drug bipartite projection for Medicare records. Fourteen graph-based methods spanning classical centrality analysis, spectral decomposition, and graph neural network architectures were evaluated and compared with five tabular one-class classification approaches. Spectral clustering in the Laplacian eigenbasis achieved the highest graph-method AUC–ROC on the credit card dataset at 0.9221, whereas a Graph Convolutional Network achieved 0.6139 on the Medicare dataset, marginally exceeding the Isolation Forest baseline of 0.6081. Graph neural networks were found to underperform spectral and classical methods on both datasets. The results indicate that the utility of graph-based fraud detection is strongly dependent on the semantic quality of the graph construction step, and that spectral methods constitute competitive baselines that are often overlooked in favor of more complex learned representations.

Index Terms—fraud detection, graph neural networks, spectral graph theory, Medicare, anomaly detection, community detection

I. INTRODUCTION

Financial fraud remains a persistent and costly problem across payment networks, health-care billing systems, insurance markets, and public benefit programs. Traditional fraud detection pipelines commonly represent each transaction, provider, or claim as an independent observation in a feature matrix. Although this representation permits the direct application of standard anomaly detection and classification algorithms, relational structure among entities is discarded. Consequently, coordinated behavior, repeated interaction patterns, and group-level irregularities may be obscured even when they are central to the mechanism by which fraud is committed.

Fraud is relational by nature. Fraud rings, coordinated billing groups, collusive providers, and repeated transaction schemes are usually expressed through relationships among individuals, institutions, accounts, prescriptions, or claims. These relationships can be represented naturally as graphs, where vertices denote entities and edges encode similarity, interaction, co-occurrence, or shared behavior. Under this

representation, suspicious behavior may appear not only as an unusual feature vector but also as an anomalous position in a network, an unusually dense community, an abnormal neighborhood, or a distinctive spectral signature.

Despite the appeal of graph-based fraud detection, much recent work has emphasized graph neural networks (GNNs) without systematically comparing them with classical graph statistics or spectral graph methods. The conditions under which graph methods outperform tabular anomaly detection remain underexplored. In particular, insufficient attention has been given to whether the edges in a constructed graph encode a domain-relevant fraud mechanism or merely reflect generic similarity in a transformed feature space.

This paper provides a comparative study of graph-theoretic fraud detection methods across two real-world datasets. First, domain-appropriate graphs were constructed from credit card transactions and Medicare provider prescribing records. Second, fourteen graph-based methods were evaluated across classical, spectral, supervised GNN, and unsupervised GNN categories. Third, all graph methods were compared with tabular machine learning baselines under matched evaluation conditions. Finally, an analysis was provided to explain why spectral methods outperformed GNN architectures in one setting and why graph construction quality was observed to be the primary determinant of graph-method utility.

The remainder of this paper is organized as follows. Related work is reviewed in Section II. The datasets and preprocessing procedures are described in Section III. Graph construction methods are presented in Section IV. Classical, spectral, and neural graph methods are described in Section V. The experimental setup and results are given in Sections VI and VII, respectively. The findings and limitations are discussed in Section VIII, and conclusions are provided in Section IX.

II. RELATED WORK

Graph-based fraud detection has been studied in settings where heterogeneous entities and interactions must be modeled jointly. Prior work has considered heterogeneous graph fraud detection, pick-based graph learning, and multi-relation graph neural networks for fraud applications [5], [6]. These approaches have shown that relationships among users, merchants, devices, claims, or providers can provide information that is not captured by isolated tabular records. However, many

such systems have been evaluated primarily within graph-learning families, and fewer studies have compared neural graph methods with classical centrality, community detection, and spectral graph baselines under the same data splits.

Several GNN architectures have become standard choices for node classification and anomaly detection. The Graph Convolutional Network (GCN) introduced by Kipf and Welling applies a first-order approximation of spectral graph convolution and has been widely adopted for semi-supervised node classification [1]. GraphSAGE was proposed to support inductive representation learning through neighborhood sampling and aggregation [2]. Graph Attention Networks (GATs) introduced learned attention weights over local neighborhoods, enabling different neighbors to contribute unequally to a node representation [3]. These architectures are often treated as natural baselines for graph-based fraud detection, although their effectiveness can depend strongly on graph topology, feature quality, and class imbalance.

Unsupervised anomaly detection on attributed networks has also been investigated. DOMINANT reconstructs both node attributes and graph structure by using a deep autoencoder and assigns anomaly scores from reconstruction errors [4]. Broader reviews of graph anomaly detection have emphasized the diversity of anomaly definitions and the importance of matching model assumptions to domain structure [7]. In fraud detection, this issue is particularly important because fraudulent behavior may appear as local irregularity, community-level coordination, or a global spectral separation.

Spectral graph theory provides a complementary perspective. The algebraic connectivity of a graph, introduced through the Fiedler value, measures the resistance of a graph to bisection and is associated with the second eigenvector of the graph Laplacian [11]. Spectral clustering uses the low-frequency eigenvectors of a graph Laplacian to identify partitions that are not necessarily visible from local degree statistics alone [12]. Because GCNs can be interpreted as simplified spectral filters, spectral methods provide a useful baseline against which learned graph filters should be compared.

Fraud detection has also been studied extensively with tabular machine learning. Credit card fraud detection benchmarks have commonly used heavily imbalanced transaction datasets and have emphasized the need for calibrated evaluation under class imbalance [13]. Medicare fraud detection has frequently been approached through tabular provider, claim, or billing features, including studies based on exclusion lists and prescribing records [8], [9]. In the present study, these research directions were connected by comparing tabular anomaly detection methods with graph-based methods on the same fraud detection tasks.

III. DATASETS AND PREPROCESSING

Two datasets were used to evaluate whether graph structure provides useful information for fraud detection. Exploratory summaries for the credit card dataset are referenced in Fig. 1, and exploratory summaries for the Medicare dataset are referenced in Fig. 2.

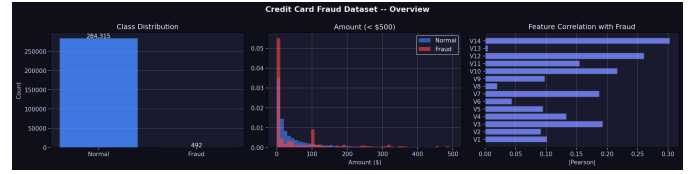


Fig. 1. Exploratory data analysis overview for the credit card fraud dataset.

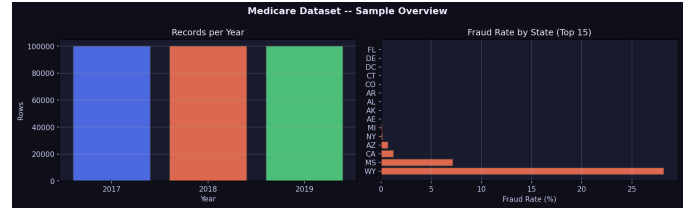


Fig. 2. Exploratory data analysis overview for the Medicare provider dataset.

A. Credit Card Fraud Dataset

The credit card dataset was the public European cardholder fraud dataset released through the ULB/Kaggle benchmark and collected in September 2013. The full dataset contains 284,807 transactions, including 492 fraudulent transactions, corresponding to a fraud rate of 0.172%. Extreme class imbalance was therefore present. The variables V_1 through V_{28} are principal components produced through a confidential transformation before public release, while Time and Amount are retained in original units. For graph construction, a Standard-Scaler transformation was applied to the 28 PCA features, and 50,000 transactions were sampled with stratification so that all 492 fraudulent transactions were preserved. The resulting graph sample had a fraud-node fraction of 0.98%.

B. Medicare Provider Fraud Dataset

The Medicare dataset was constructed from CMS Medicare Part D prescribing files for 2017, 2018, and 2019. Fraud labels were obtained from the Office of Inspector General List of Excluded Individuals and Entities (LEIE). The raw data consisted of approximately 11.9 GB across three yearly CSV files. A memory-constrained sample of 6 million rows was used, with 2 million rows drawn from each year and stratification applied by provider fraud status. After deduplication, 7,701 unique providers remained, of whom 137 were labeled as fraudulent, corresponding to a fraud rate of 1.78%. The OIG LEIE contained 8,306 excluded National Provider Identifiers nationally and was used as the fraud ground truth.

IV. GRAPH CONSTRUCTION

Two graph construction strategies were used. A similarity graph was constructed for credit card transactions, whereas a domain-driven bipartite projection was constructed for Medicare provider records. Visual summaries of the two construction procedures are referenced in Fig. 3 and Fig. 4.

A. Credit Card K -Nearest-Neighbor Similarity Graph

For the credit card data, each sampled transaction was represented as a node. Edges were constructed through cosine similarity in the 28-dimensional standardized PCA space. The symmetric k -nearest-neighbor adjacency matrix was defined as shown in Eq. (1), with $k = 10$:

$$A_{ij} = \begin{cases} 1, & \text{if } j \in \text{kNN}(i) \text{ or } i \in \text{kNN}(j), \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The resulting undirected graph contained $|V| = 50,000$ nodes and $|E| = 347,127$ edges. Although this construction provided a tractable graph over transactions, it encoded proximity in a compressed PCA representation rather than direct behavioral or transactional relations. Therefore, the semantic meaning of the graph structure was expected to be weaker than in a graph whose edges correspond to known fraud mechanisms.

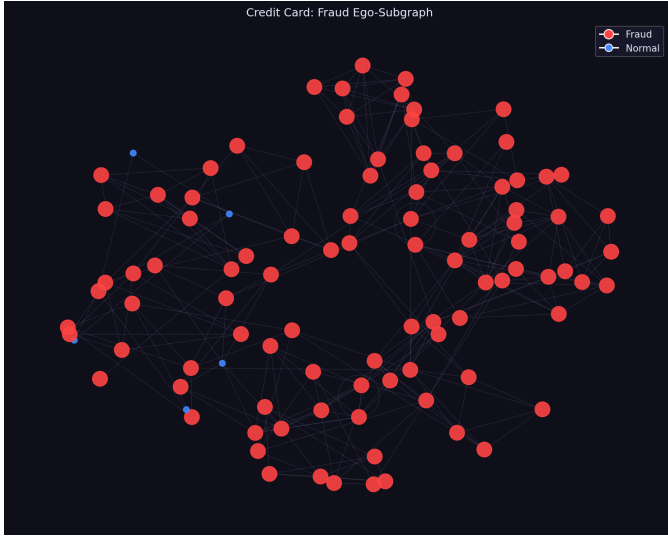


Fig. 3. Illustrative subgraph of the credit card k -nearest-neighbor graph.

B. Medicare Provider–Drug Bipartite Projection

For the Medicare data, a two-stage construction was used. First, a bipartite graph $B = (V_P \cup V_D, E_B)$ was formed, where V_P denotes provider nodes and V_D denotes drug nodes. An edge $(p, d) \in E_B$ was inserted when provider p prescribed drug d in the combined 2017–2019 records. Second, a unipartite provider graph $G = (V_P, E_G)$ was generated by projecting the bipartite graph onto providers. The projection rule was defined as shown in Eq. (2):

$$(p_1, p_2) \in E_G \iff |D(p_1) \cap D(p_2)| \geq \tau, \quad \tau = 5. \quad (2)$$

Drugs prescribed by more than 25% of all providers were removed before projection. Without this hub-drug removal, the projected graph would have contained approximately 334 million edges and would have approached a near-complete graph, thereby eliminating useful structural signal. Neighborhood size was additionally capped at $K = 25$ per provider to limit memory consumption. The resulting graph contained

$|V| = 7,701$ nodes and $|E| = 113,350$ edges. In contrast to the credit card graph, these edges encoded shared prescribing patterns, which can be interpreted as a domain-relevant signal for coordinated or anomalous provider behavior.

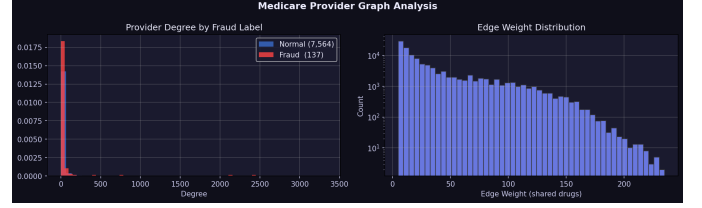


Fig. 4. Medicare graph construction through provider–drug bipartite modeling and provider–provider projection.

V. METHODS

Three families of graph methods were evaluated: classical graph analysis, spectral graph theory, and graph neural networks. The classical and spectral methods were included to test whether simpler structural baselines could compete with learned graph representations.

A. Classical Graph Analysis

Six centrality and local structure measures were computed for all nodes. Degree centrality was computed as shown in Eq. (3), where $k(v)$ denotes the degree of node v :

$$C_D(v) = \frac{k(v)}{|V| - 1}. \quad (3)$$

PageRank was computed with damping parameter $\alpha = 0.85$ for 200 iterations according to Eq. (4):

$$PR(v) = \frac{1 - \alpha}{|V|} + \alpha \sum_{u \in \mathcal{N}(v)} \frac{PR(u)}{k(u)}. \quad (4)$$

Betweenness centrality was approximated by sampling $k = 300$ pivot nodes for large graphs and was defined as shown in Eq. (5):

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma(s, t | v)}{\sigma(s, t)}. \quad (5)$$

Closeness centrality was computed as in Eq. (6):

$$C_C(v) = \frac{|V| - 1}{\sum_{u \neq v} d(u, v)}. \quad (6)$$

Eigenvector centrality was given by the leading eigenvector of the adjacency matrix A . Local clustering coefficient was computed according to Eq. (7):

$$C(v) = \frac{2|\{(u, w) : u, w \in \mathcal{N}(v), (u, w) \in E\}|}{k(v)(k(v) - 1)}. \quad (7)$$

The seven-dimensional feature vector formed by these six measures plus raw degree was used as input to a Random Forest classifier trained with balanced class weights and five-fold cross-validation. The resulting centrality distributions are referenced in Fig. 5 and Fig. 6.



Fig. 5. Centrality distributions for fraudulent and legitimate credit card transaction nodes.

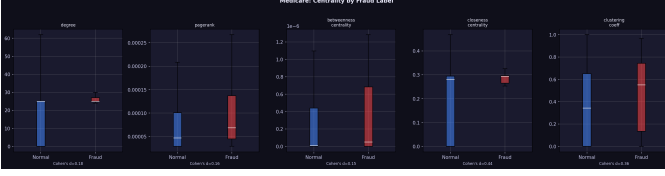


Fig. 6. Centrality distributions for fraudulent and legitimate Medicare provider nodes.

Community detection was performed with the Louvain algorithm. Modularity was maximized according to Eq. (8), where m is the number of edges, k_i is the degree of node i , and $\delta(c_i, c_j)$ indicates whether nodes i and j are assigned to the same community:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j). \quad (8)$$

Community-level fraud rates were then computed for interpretation, and the largest communities are referenced in Fig. 7.

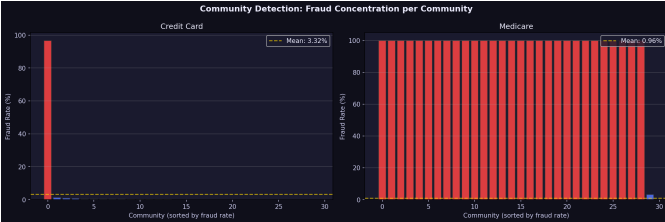


Fig. 7. Fraud rates by Louvain community for the largest detected communities.

B. Spectral Graph Theory

The normalized graph Laplacian was used to characterize global graph structure. It was defined as shown in Eq. (9):

$$\mathcal{L} = D^{-1/2}(D - A)D^{-1/2}. \quad (9)$$

The eigendecomposition in Eq. (10) was then computed, with eigenvalues ordered as $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$:

$$\mathcal{L} = U\Lambda U^T. \quad (10)$$

The second smallest eigenvalue λ_2 , also known as the Fiedler value or algebraic connectivity, measures resistance to graph bisection. The corresponding eigenvector u_2 provides a low-dimensional direction that was used directly as a node anomaly score. The eigenvalue spectra and Fiedler-based anomaly score summaries are referenced in Fig. 8, Fig. 9, and Fig. 10.

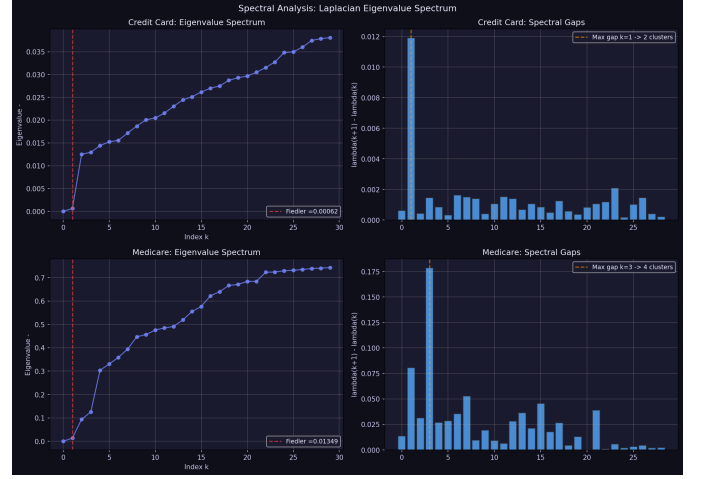


Fig. 8. Laplacian eigenvalue spectra and spectral gaps for the credit card and Medicare graphs.

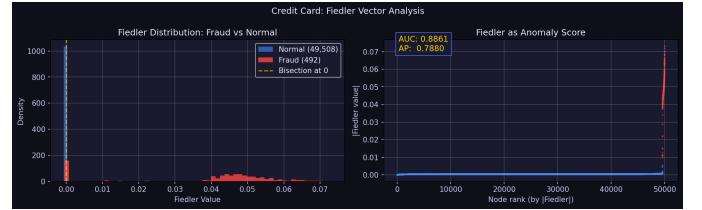


Fig. 9. Credit card Fiedler vector distribution and ROC analysis.

Spectral clustering was performed by computing the bottom k eigenvectors of $\mathcal{L}$ with $k = 10$ and applying K-means in the resulting spectral embedding. Node anomaly scores were assigned as the empirical fraud rate of each cluster. The relationship between spectral graph theory and GCN propagation was also considered. Spectral graph convolution can be written as shown in Eq. (11):

$$g_\theta * x = U g_\theta(\Lambda) U^T x. \quad (11)$$

This operation can be approximated by degree- K Chebyshev polynomials as shown in Eq. (12):

$$g_\theta * x \approx \sum_{k=0}^K \theta_k T_k(\tilde{L})x. \quad (12)$$

Setting $K = 1$ and $\lambda_{\max} = 2$ yields the GCN propagation rule in Eq. (13), where $\tilde{A} = A + I$ and $\tilde{D}$ is the corresponding

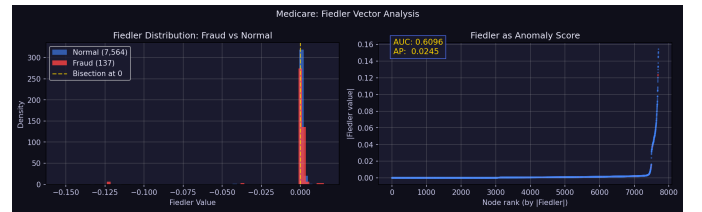


Fig. 10. Medicare Fiedler vector distribution and ROC analysis.

degree matrix:

$$H^{(l+1)} = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)}). \quad (13)$$

Thus, the GCN can be interpreted as a first-order spectral filter [1].

C. Graph Neural Networks

All GNN models were implemented using PyTorch Geometric 2.7. Input features were the seven centrality measures described above. Binary cross-entropy loss with class weighting $w_+ = n_{\text{neg}}/n_{\text{pos}}$ was used throughout, and the Adam optimizer with weight decay 5×10^{-4} was applied.

The GCN model used two GCNConv layers with dimensions $\text{input} \rightarrow 64 \rightarrow 32$, ReLU activations, dropout with $p = 0.5$, and a linear output layer followed by a sigmoid. The learning rate was set to 10^{-2} , and models were trained for 200 epochs. The GraphSAGE model used mean aggregation with self-feature concatenation and the same hidden dimensions and training protocol as the GCN. The GAT model used four attention heads with eight units per head. Attention coefficients were computed as shown in Eq. (14):

$$\alpha_{ij} = \text{softmax}_j(\text{LeakyReLU}(a^T [W h_i \parallel W h_j])). \quad (14)$$

The GAT learning rate was set to 5×10^{-3} , and training was performed for 200 epochs.

DOMINANT was evaluated as an unsupervised graph anomaly detection method [4]. Node anomaly scores were computed from joint reconstruction error over graph structure and node attributes as shown in Eq. (15):

$$s(v) = \alpha \|a_v - \hat{a}_v\|^2 + (1 - \alpha) \|x_v - \hat{x}_v\|^2, \quad \alpha = 0.5. \quad (15)$$

The model was trained for 100 epochs with hidden dimension 64. Training behavior for the supervised GNN models is referenced in Fig. 11.

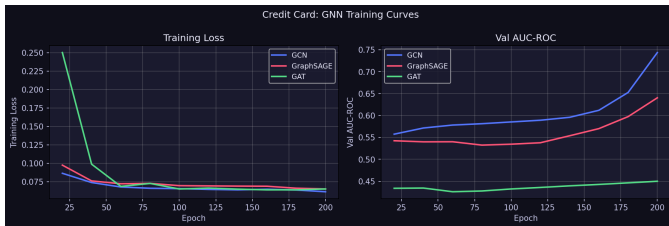


Fig. 11. Training loss and validation AUC curves for supervised GNN models on the credit card graph.

VI. EXPERIMENTAL SETUP

All experiments used stratified train, validation, and test splits with proportions of 70%, 10%, and 20%, respectively. The credit card split contained 34,999 training nodes, 4,999 validation nodes, and 10,002 test nodes. The Medicare split contained 5,389 training nodes, 769 validation nodes, and 1,543 test nodes. AUC-ROC was used as the primary metric, while AUC-PRC, also reported as Average Precision, was used as a secondary metric. For GNN models, F1 score at

an optimal threshold was additionally tracked during model selection, although the comparative tables focus on AUC-ROC and AUC-PRC.

Tabular machine learning baselines were obtained from a companion anomaly detection study conducted on the same datasets under the same preprocessing and split conditions. The baseline methods included one-class support vector machines (OCSVM), Gaussian mixture models (GMM), Isolation Forest, Local Outlier Factor (LOF), and an ensemble anomaly detector. These baselines were used to determine whether graph construction and graph-based modeling provided measurable gains beyond conventional tabular anomaly detection.

VII. RESULTS

The full comparison across tabular, classical graph, spectral graph, supervised GNN, and unsupervised GNN methods is summarized in Table I. A visual summary of the same method families is referenced in Fig. 12. On the credit card dataset, the strongest overall AUC-ROC was obtained by the tabular GMM baseline at 0.9588. The best graph-based AUC-ROC was obtained by spectral clustering at 0.9221, corresponding to a 3.83% relative gap from the strongest tabular baseline. The Fiedler anomaly score produced an AUC-ROC of 0.8861 and the highest AUC-PRC among the listed methods at 0.7880, indicating that spectral information remained useful under severe class imbalance.

On the Medicare dataset, the best AUC-ROC was obtained by the GCN model at 0.6139. This result exceeded the strongest tabular baseline, Isolation Forest, which achieved 0.6081, by 0.0058 AUC-ROC, or approximately 0.95% in relative terms. The Fiedler anomaly score achieved 0.6096 AUC-ROC, suggesting that spectral structure was nearly as competitive as the best neural model. However, the absolute improvement over tabular baselines was marginal, and the practical significance of such a small gain would require validation under deployment-specific operating constraints.

GNN training times are reported in Table II. DOMINANT required 1,263.7 seconds on the credit card graph, or approximately 21 minutes, which was substantially longer than the supervised GNN models. The GCN required 35.2 seconds on credit card data and 9.7 seconds on Medicare data, while GraphSAGE and GAT showed intermediate computational profiles. The high cost of DOMINANT on the larger graph was attributed to joint reconstruction of graph structure and attributes.

Community analysis provided additional interpretability. In the credit card graph, Louvain community 0 contained 396 nodes, of which 383 were fraudulent, corresponding to a 96.7% internal fraud rate. This result suggested that at least some fraudulent transactions were concentrated in highly enriched graph regions even though the overall k -nearest-neighbor construction was not semantically grounded. The Fiedler values were also informative: the credit card graph had $\lambda_2 = 0.000619$, whereas the Medicare graph had $\lambda_2 = 0.013493$. The smaller credit card Fiedler value indicated

TABLE I
PERFORMANCE COMPARISON OF ALL EVALUATED METHODS ON BOTH DATASETS. AUC-ROC AND AUC-PRC ARE REPORTED. BEST RESULT PER DATASET IS SHOWN IN BOLD.

Dataset	Category	Method	AUC-ROC	AUC-PRC
Credit Card	Tabular-ML	GMM	0.9588	0.5720
Credit Card	Tabular-ML	LOF	0.9532	0.6391
Credit Card	Tabular-ML	Ensemble	0.9535	0.5191
Credit Card	Tabular-ML	OCSVM	0.9514	0.2731
Credit Card	Tabular-ML	Isolation Forest	0.9496	0.1543
Credit Card	Graph-Spectral	Spectral Clustering ($k = 10$)	0.9221	0.7489
Credit Card	Graph-Spectral	Fiedler Anomaly Score	0.8861	0.7880
Credit Card	Graph-Classical	RF on Centrality	0.8839	0.4431
Credit Card	Graph-GNN	GCN	0.6144	0.0693
Credit Card	Graph-GNN-Unsup	DOMINANT	0.5708	0.0197
Credit Card	Graph-GNN	GraphSAGE	0.5171	0.0643
Credit Card	Graph-GNN	GAT	0.3586	0.0136
Medicare	Graph-GNN	GCN	0.6139	0.0273
Medicare	Graph-Spectral	Fiedler Anomaly Score	0.6096	0.0245
Medicare	Tabular-ML	Isolation Forest	0.6081	0.0038
Medicare	Tabular-ML	LOF	0.6043	0.0021
Medicare	Tabular-ML	GMM	0.5977	0.0023
Medicare	Tabular-ML	Ensemble	0.5974	0.0037
Medicare	Graph-Spectral	Spectral Clustering ($k = 10$)	0.5779	0.0251
Medicare	Graph-GNN-Unsup	DOMINANT	0.5509	0.0200
Medicare	Tabular-ML	OCSVM	0.5261	0.0019
Medicare	Graph-GNN	GAT	0.4361	0.0159
Medicare	Graph-Classical	RF on Centrality	0.4332	0.0144
Medicare	Graph-GNN	GraphSAGE	0.4309	0.0171

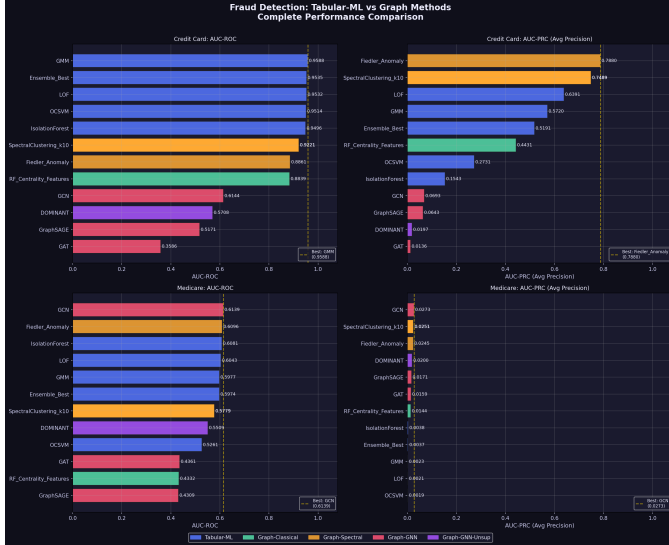


Fig. 12. Master comparison of all evaluated methods across both datasets and primary metrics.

TABLE II
GNN TRAINING TIME BY METHOD AND DATASET.

Method	Credit Card (s)	Medicare (s)
GCN	35.2	9.7
GraphSAGE	28.4	7.7
GAT	51.8	16.5
DOMINANT	1263.7	37.4

weaker algebraic connectivity and greater susceptibility to graph partitioning.

VIII. DISCUSSION

Spectral methods substantially outperformed all GNN architectures on the credit card dataset. This result can be explained by the difference between global topology exploitation and local neighborhood aggregation. Spectral clustering used low-frequency Laplacian eigenvectors that summarized broad partition structure across the graph. In contrast, GNN models aggregated local neighborhoods and were trained under extreme class imbalance, causing minority fraud labels to be weakly represented during supervised learning. In addition, the GNN input features were centrality vectors rather than raw transaction features, which restricted the information available to learned message passing. Under these conditions, additional neural complexity did not translate into stronger fraud discrimination.

Graph construction quality was observed to be the primary determinant of graph-method utility. The Medicare graph encoded shared prescribing behavior through a provider-drug projection, and such edges were more closely aligned with plausible collusion or coordinated billing mechanisms. The credit card graph, by contrast, encoded cosine proximity in PCA space. Since the released credit card features were already transformed for confidentiality, the constructed edges did not necessarily correspond to meaningful interactions among cardholders, merchants, or devices. This distinction explains why a GCN could marginally surpass tabular baselines on Medicare, whereas credit card graph methods remained below the strongest tabular models despite strong spectral performance.

The relative success of GCN over GraphSAGE and GAT on Medicare is also consistent with the topology of the provider graph. Provider networks often contain many low-degree peripheral nodes and a smaller number of high-degree hubs. GraphSAGE relies on inductive neighborhood sampling, which may miss informative structure when relevant signals are sparse or unevenly distributed. GAT introduces attention mechanisms that may become unstable or poorly calibrated when neighborhoods are highly heterogeneous and labels are scarce. The normalized aggregation used by GCN automatically downweights high-degree hubs through degree normalization, making it better suited to the observed scale-free provider topology.

The practical implication is that graph methods should be deployed for fraud detection when the edge definition encodes a known fraud mechanism. Provider co-prescribing, shared devices, common addresses, coordinated claims, or repeated account interactions can produce edges with direct domain meaning. Generic similarity graphs, especially those built over compressed or anonymized feature spaces, are less likely to provide a durable advantage over tabular models. Therefore, graph construction should be treated as a modeling decision of primary importance rather than a preprocessing detail.

A. Limitations

Several limitations should be noted. The public credit card dataset had been transformed by PCA before release, so the original transaction attributes and possible entity relationships were unavailable. The Medicare analysis was constrained by memory and used 6 million of approximately 60 million rows; hub-drug removal and the $K = 25$ neighborhood cap may have discarded informative edges. GNN experiments were conducted with a single random seed, and variance across repeated trials was not estimated. Tabular baselines were taken from a companion project and were not rerun within the graph codebase. Finally, the Medicare improvement of approximately 0.95% over the strongest tabular baseline was marginal and would likely fall below the practical significance threshold required in many deployed fraud detection systems.

IX. CONCLUSION

Graph-theoretic methods were found to provide competitive but not uniformly superior performance relative to tabular anomaly detection. On the credit card dataset, spectral clustering achieved the strongest graph-based AUC-ROC at 0.9221, although the tabular GMM baseline achieved a higher score of 0.9588. On the Medicare dataset, the GCN achieved the best AUC-ROC at 0.6139 and marginally exceeded the strongest tabular baseline. Across both datasets, GNNs generally underperformed simpler spectral and classical methods, indicating that neural graph models should not be assumed to dominate well-designed graph baselines.

The central finding is that graph construction quality determines whether graph-based fraud detection offers an advantage. When edges encode domain-relevant relationships, as in the Medicare provider-drug projection, graph methods

may provide incremental value. When edges represent generic similarity in transformed feature space, as in the credit card k -nearest-neighbor graph, the benefits are more limited. Future work should evaluate richer node features, including raw transaction attributes where available, heterogeneous graph construction for Medicare, and temporal graph models capable of representing recurring fraud patterns over time.

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